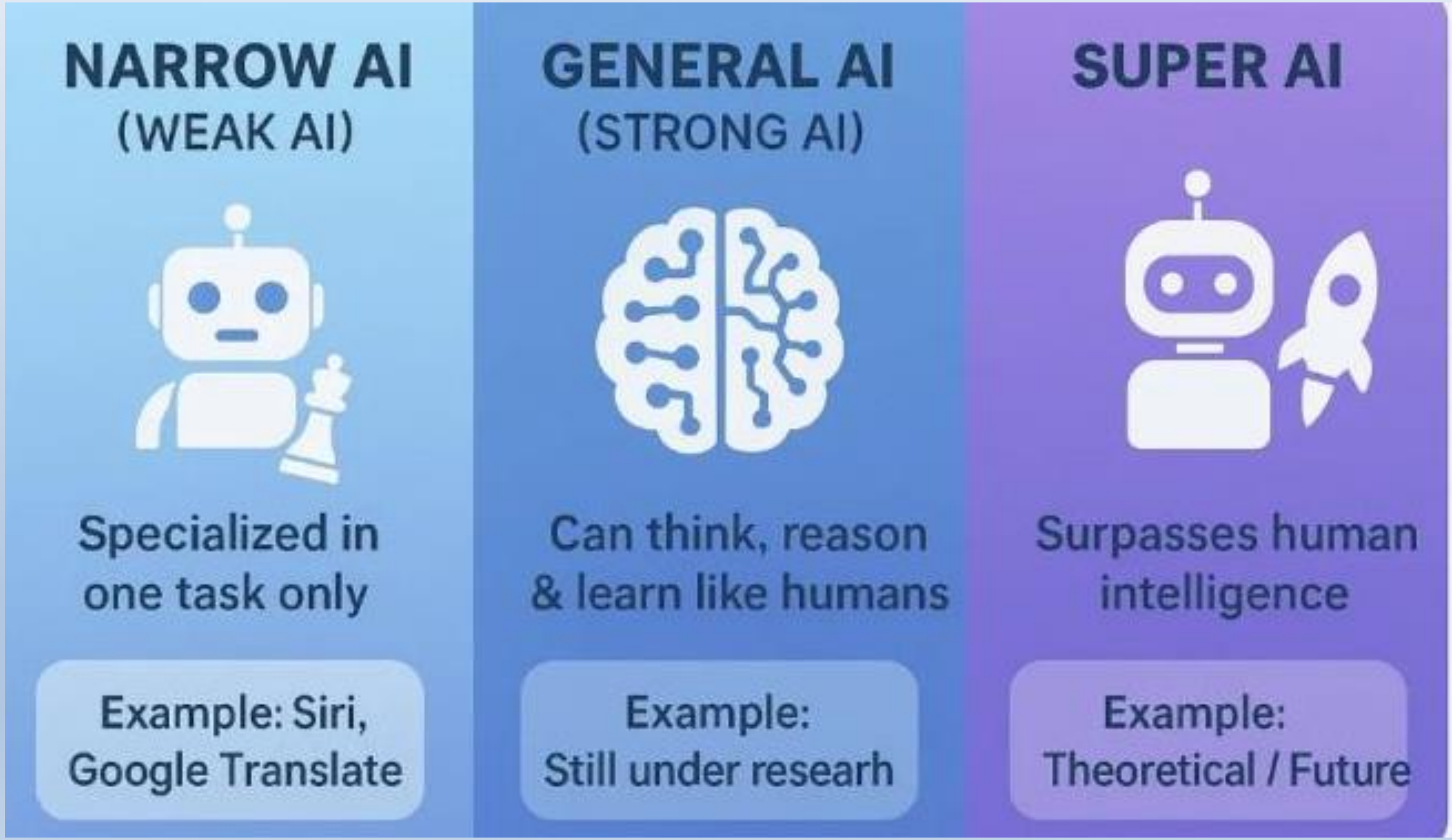
INTRODUCTION TO AI

Evolution of AI from Simple Concepts

Evolution of AI

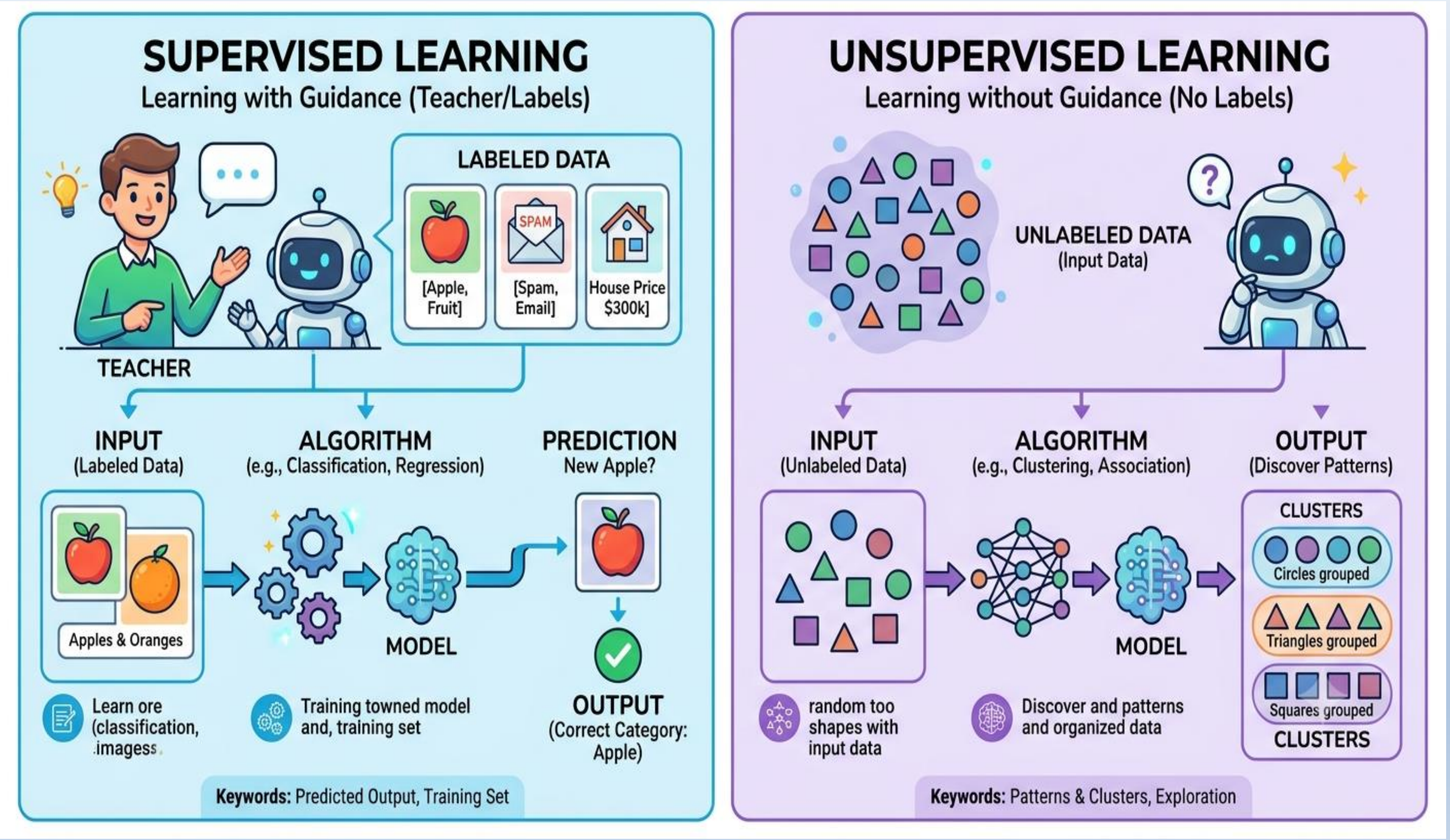
- 1950s: Birth of AI
- Machine Learning
- Deep Learning
- Generative AI

Types of AI



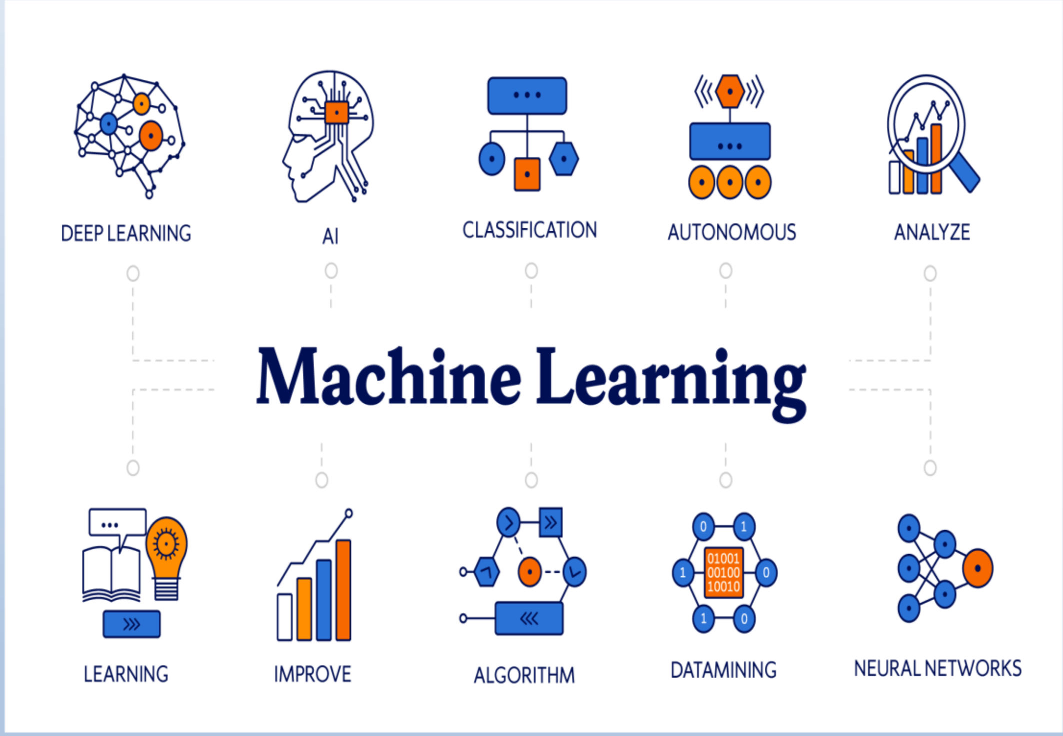
NARROW AI
(WEAK AI)
Specialized in one task only
Example: Siri, Google Translate
GENERAL AI
(STRONG AI)
Can think, reason & learn like humans
Example: Still under researh
SUPER AI
Surpasses human intelligence
Example: Theoretical / Future

Supervised vs Unsupervised



Machine Learning

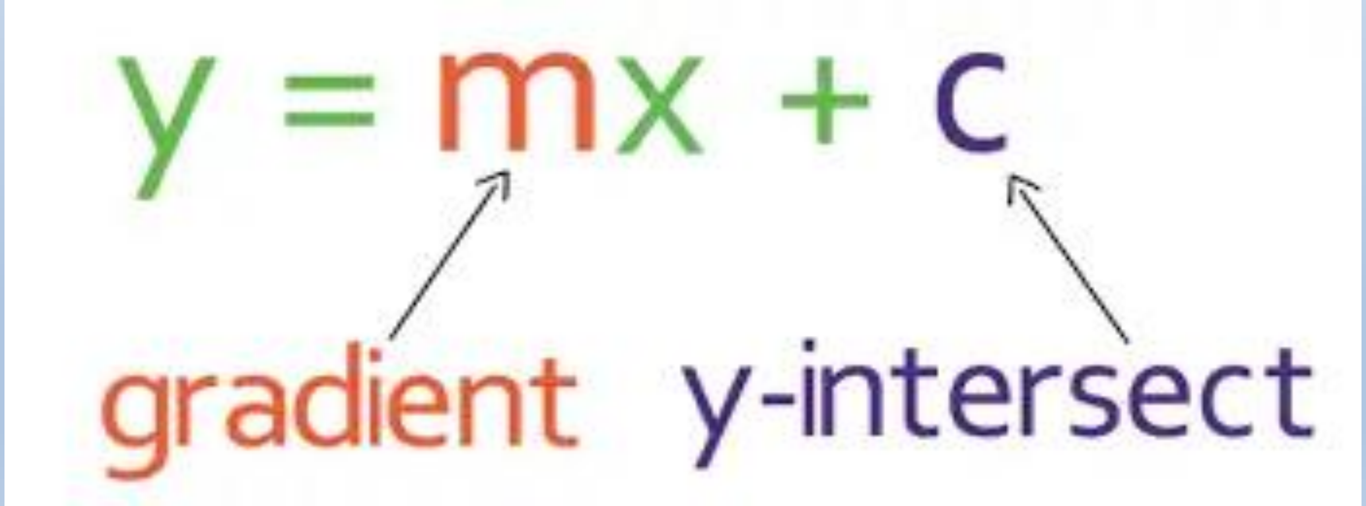
- Learns from Data
- Finds Patterns
- Makes Predictions
- Improves Accuracy
- Automates Tasks
- Solves Problems



Linear Regression

- Equation: $y = mx + c$

- Predicts continuous values



Linear Regression Example

X=[1,2,3,4,5] y=[3,5,7,9,11]

Numerator:

$\sum(X - Xmean)\cdot(y - ymean)$

Denominator:

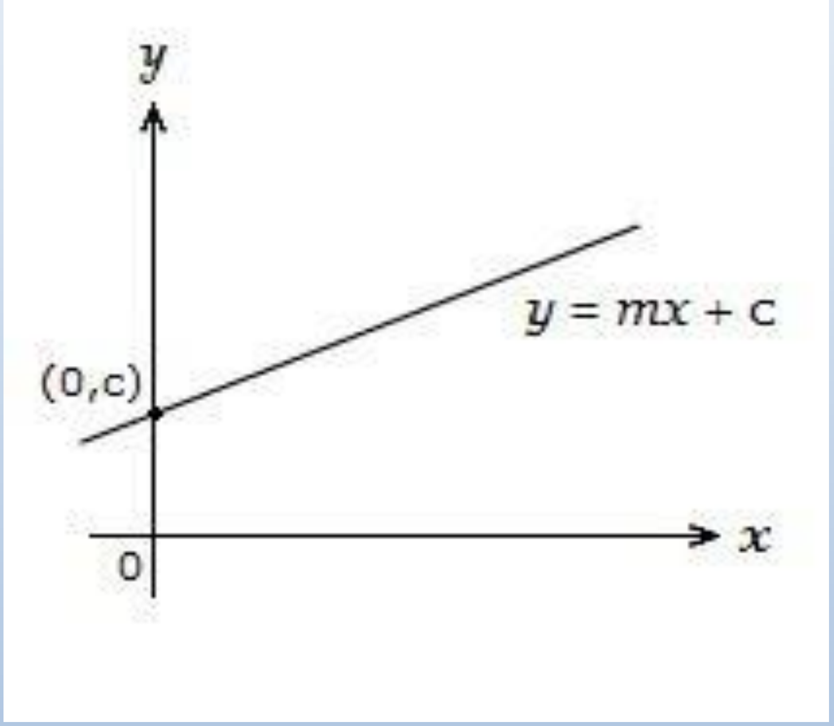
$\sum(X - Xmean)2$

slope (m):

m=Numerator/Denominator

intercept (c):

c=ymean−m·Xmean



Result: y=2x+1

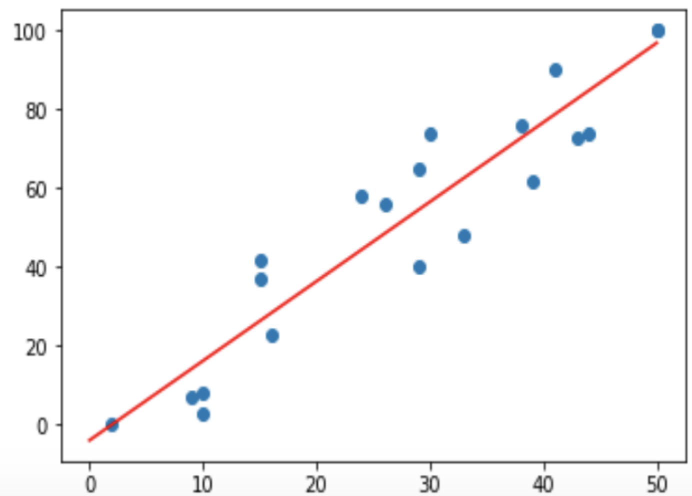
Python Experiment

- Train Linear Regression
- Predict outputs
- Plot regression line

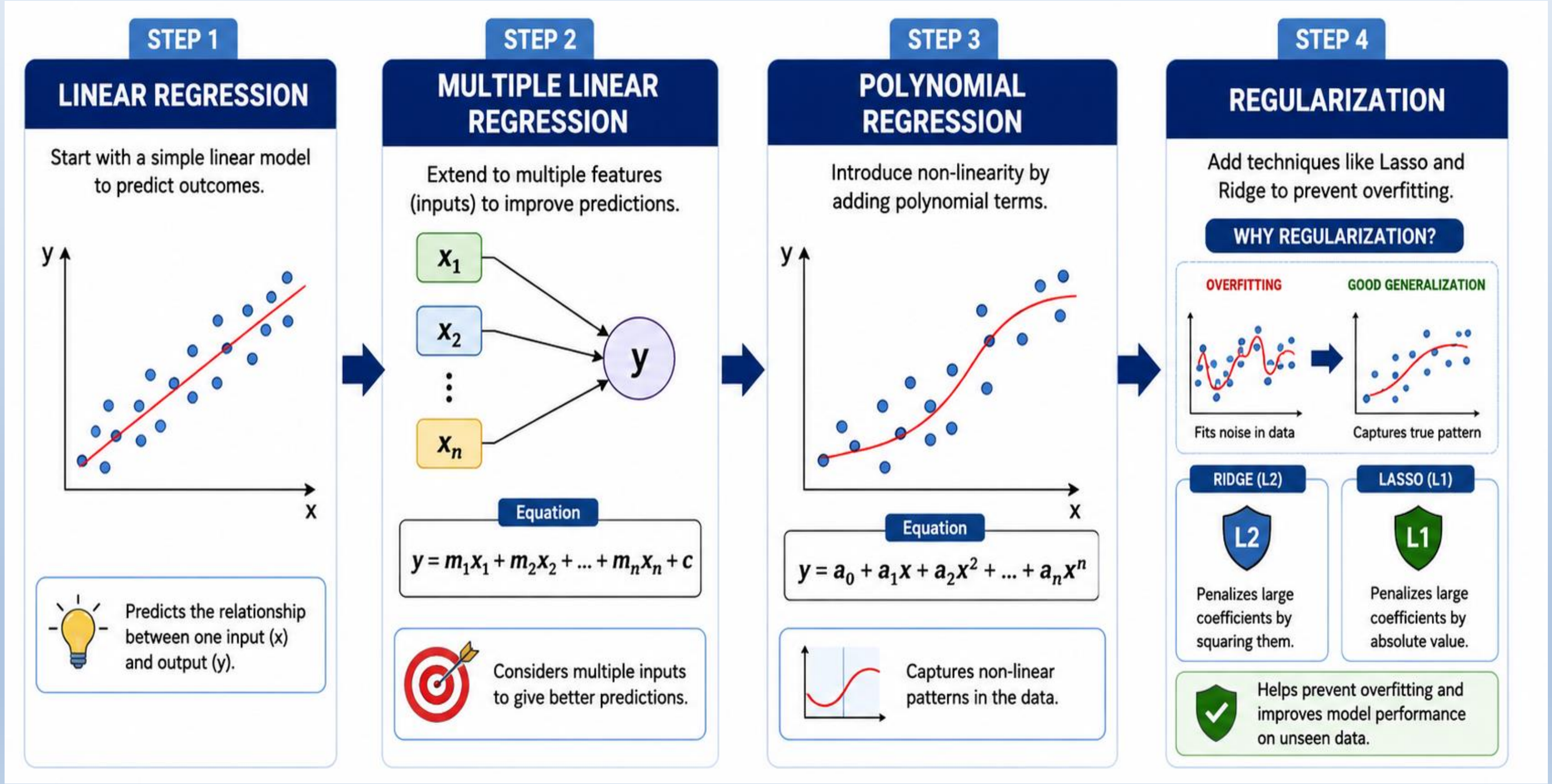
Step #6 - Dataviz

In [12]:
```
x_lin_reg = range(0, 51)
y_lin_reg = predict(x_lin_reg)
plt.scatter(x, y)
plt.plot(x_lin_reg, y_lin_reg, c = 'r')
```

Out[12]: [<matplotlib.lines.Line2D at 0x1a19bff550>]

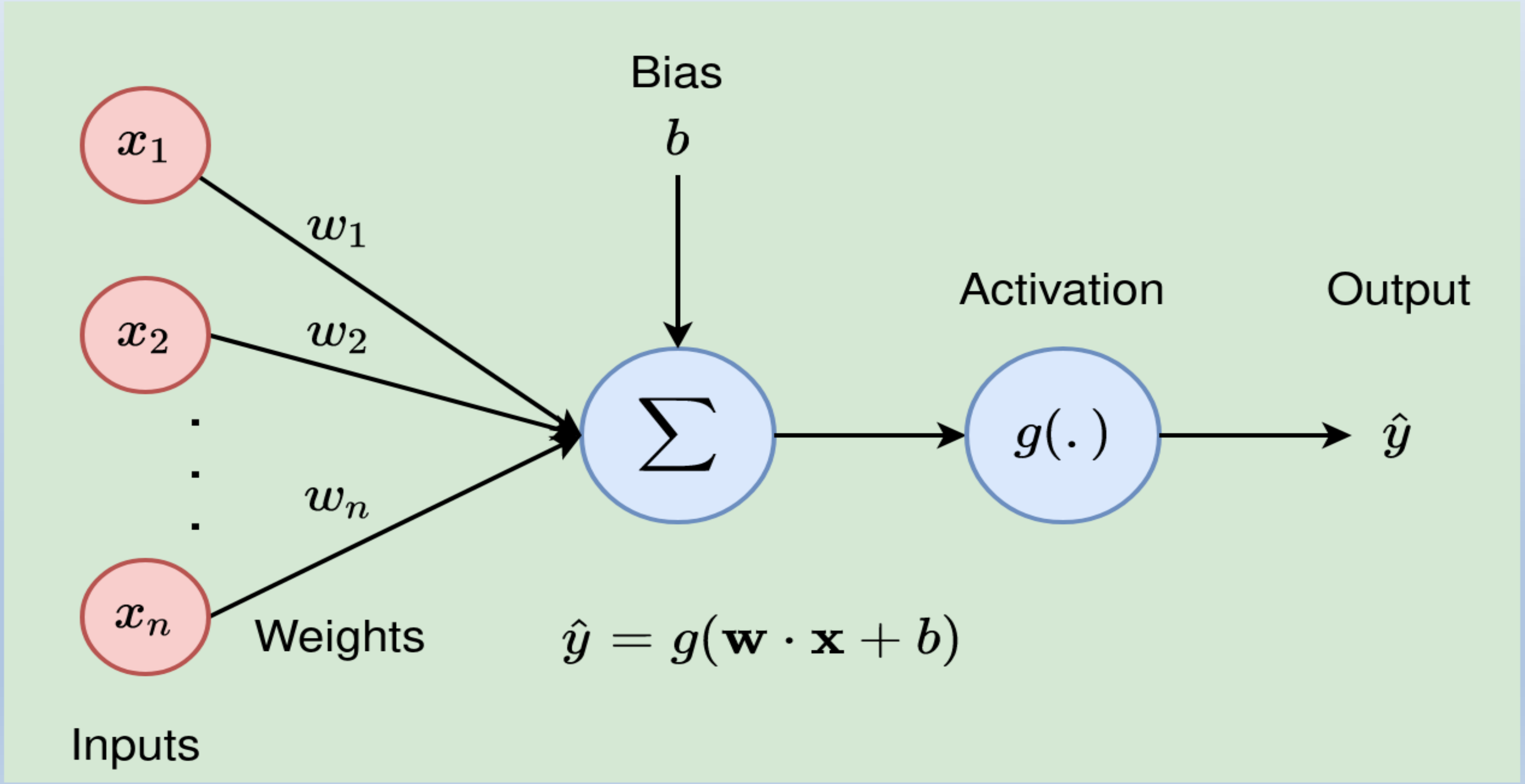


From Regression to ML



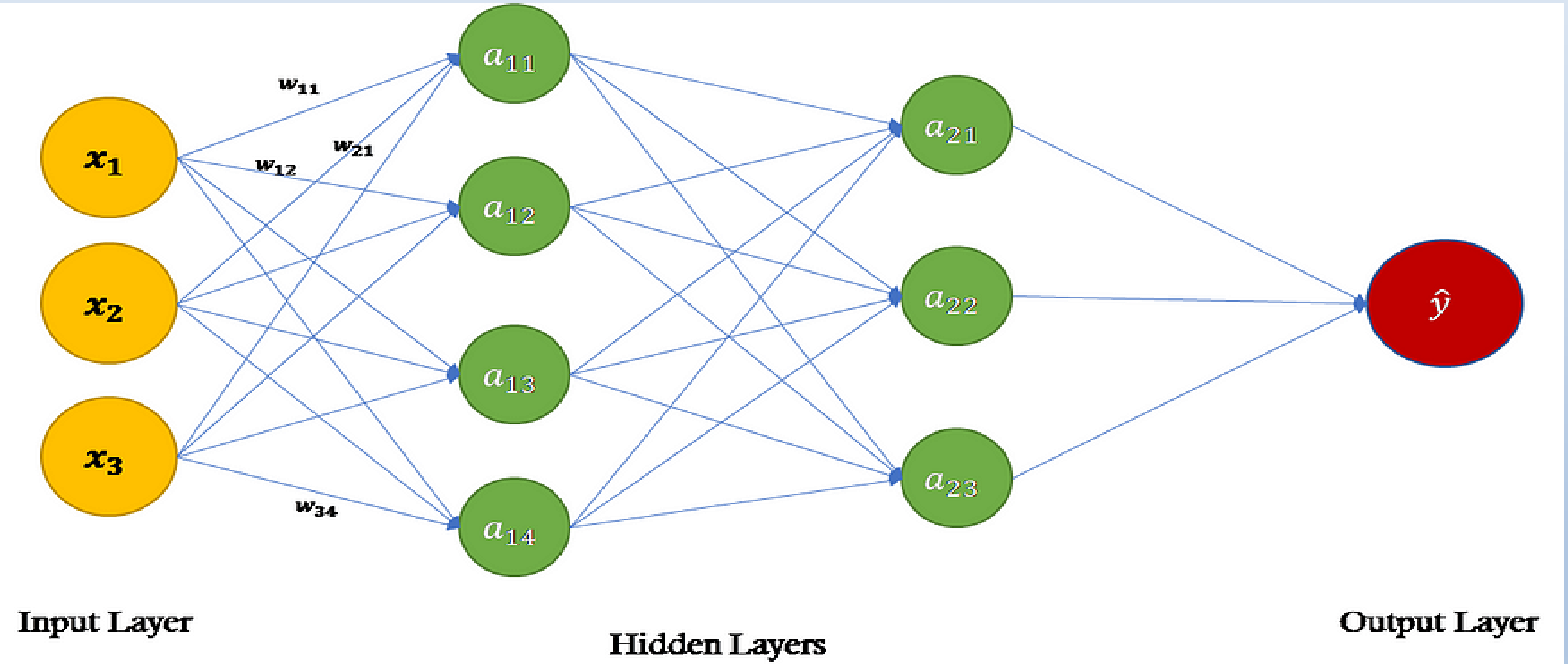
Artificial Neural Networks

Single Neuron (Perceptron):



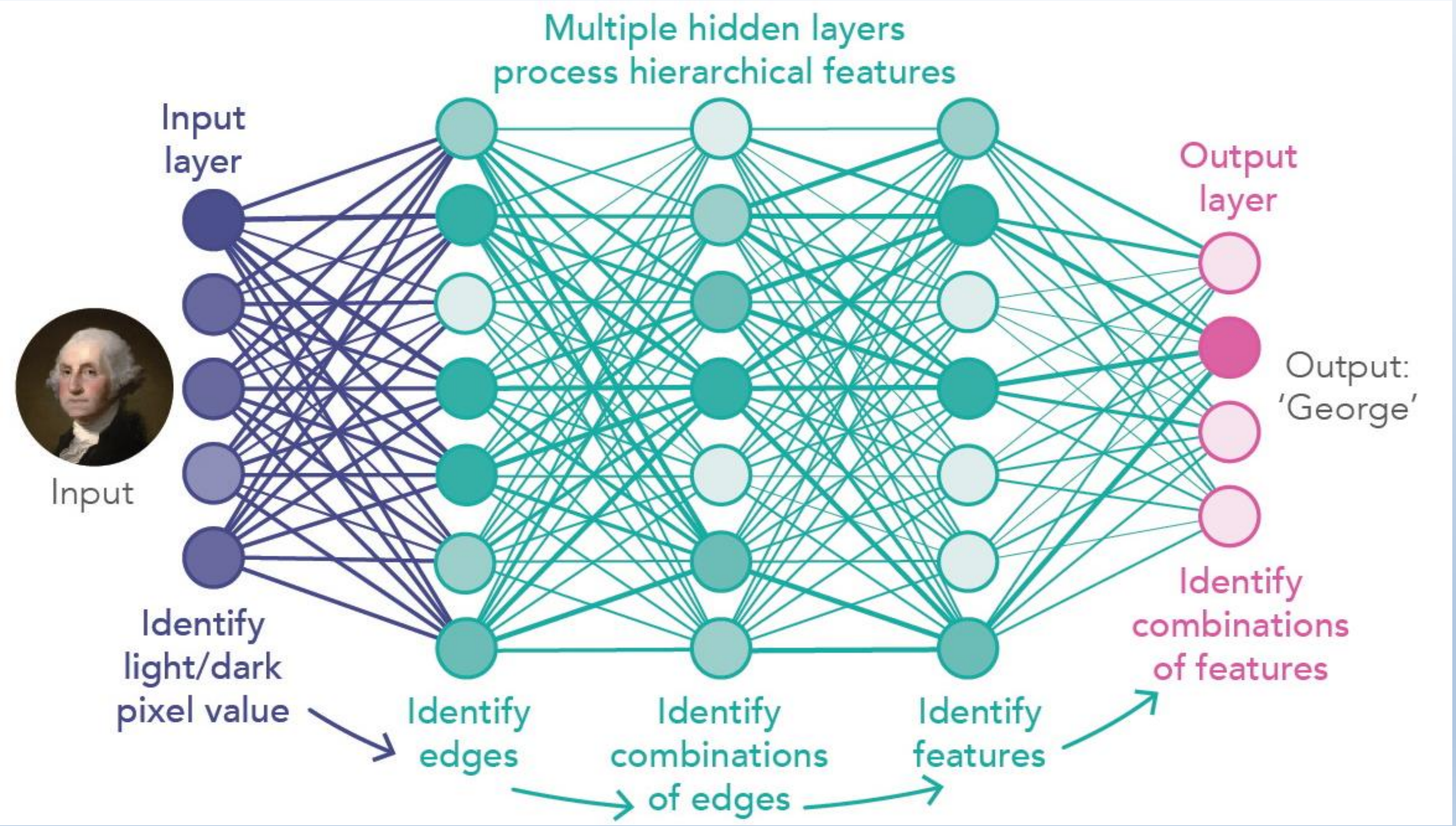
Artificial Neural Networks

Multi-Layer Perceptron (MLP):

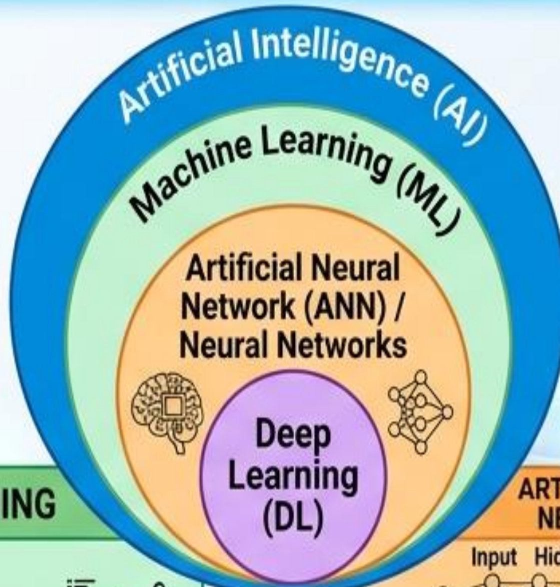


Multilayer Perceptron

Deep Learning



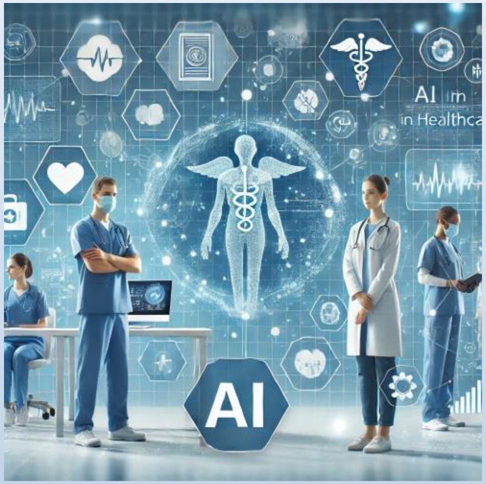
ML vs ANN vs Deep Learning



CHARACTERISTICS	MACHINE LEARNING	ARTIFICIAL NEURAL NETWORKS (ANN)	DEEP LEARNING (DL)
Visual Structure	Model	Input Hidden Output Input Shallow Structure	Deep Structure (Multi-layer)
Data Requirements	Moderate (e.g., thousands of examples)	More data (e.g., tens of thousands)	Massive (e.g., millions, petabytes)
Feature Engineering	Manual by domain experts	Some internal feature discovery, but often pre-engineered	Automatic (Feature extraction learned end-to-end)
Interpretability Examples	Spam detection, basic recommendation, fraud detection	Pattern recognition, function approximation	Image recognition (e.g., self-driving car visual), natural language processing, complex gaming

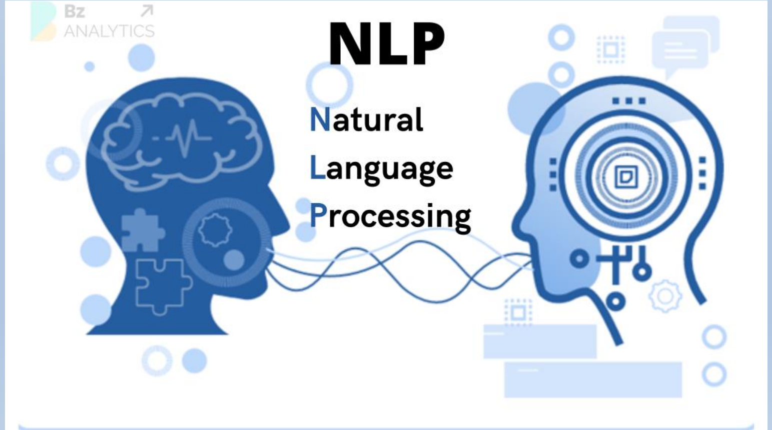
Increased Complexity & Data

AI Applications

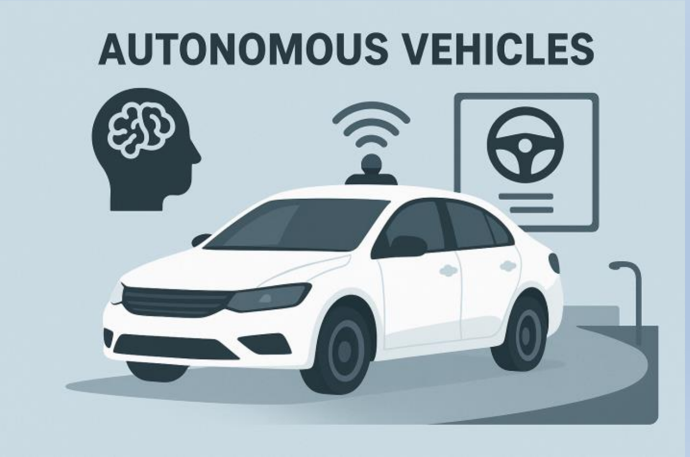


Healthcare

Finance



NLP



Autonomous Vehicles

Conclusion

- Machine Learning helps AI learn from data and make better decisions.
- It is used in many fields like healthcare, finance, education, and transportation.
- Keep exploring Machine Learning to understand how it is shaping the future of AI.